

Hunters Hill High School

Extension 2, Mathematics

Trial Examination, 2018



Hunters Hill
High School

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- The marks for each question are shown on the paper
- Show relevant mathematical reasoning and/or calculations in Questions 11-16

Total Marks: 100

Section I Page 3 – 6
10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II Pages 7 – 13
90 marks

- Attempt Questions 11-16
- Begin a **new sheet** for each question
- Allow about 2 hour 45 minutes for this section

Section I

Total marks – 10

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1–10.

1. Which of the following ellipses has focus $(1, 0)$ and directrix $x = 4$?

(A) $\frac{x^2}{3} + \frac{y^2}{4} = 1$

(B) $4x^2 + 9y^2 = 36$

(C) $3x^2 + 4y^2 = 12$

(D) $\frac{x^2}{4} + \frac{y^2}{\sqrt{3}} = 1$

2. Find $\text{Arg } z_1$ where $z_1 = (-1 + \sqrt{3}i)^8$

(A) $\frac{16\pi}{3}$

(B) $\frac{2\pi}{3}$

(C) $-\frac{2\pi}{3}$

(D) $-\frac{\pi}{3}$

3. $P(x)$ is a polynomial of degree 4 with real coefficients. $P(x)$ has $x = 2$ as a root of multiplicity 2, and $x = -i$ as a root.

Which of the following expressions is a factorised form of $P(x)$ over the complex numbers?

(A) $P(x) = (x - 2)^2(x - 1)(x + 1)$

(B) $P(x) = (x + 2)^2(x - 1)^2$

(C) $P(x) = (x - 2)^2(x - i)(x + i)$

(D) $P(x) = (x + 2)^2(x - i)(x + i)$

4. The points $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the chord PQ subtends a right angle at $(0, 0)$.

Which of the following expressions is correct?

- (A) $\tan \theta \cdot \tan \phi = \frac{a^2}{b^2}$
 (B) $\tan \theta \cdot \tan \phi = -\frac{a^2}{b^2}$
 (C) $\tan \theta \cdot \tan \phi = \frac{b^2}{a^2}$
 (D) $\tan \theta \cdot \tan \phi = -\frac{b^2}{a^2}$

5. Find $\int x \ln x \, dx$

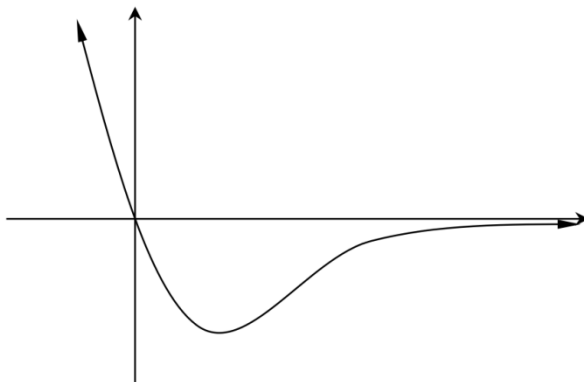
- (A) $1 - x \ln x + c$
 (B) $1 - \int \ln x \, dx + c$
 (C) $\frac{1}{2}x^2 \ln x - \frac{1}{4}x^2 + c$
 (D) $\frac{1}{2}x^2 \ln x - \frac{1}{4}x^2 + c$

6. A particle of mass m is undergoing circular motion in a circle of radius r with angular velocity ω . Let the particle's tangential velocity be V and let its tangential acceleration be a .

Which of the following expressions is correct?

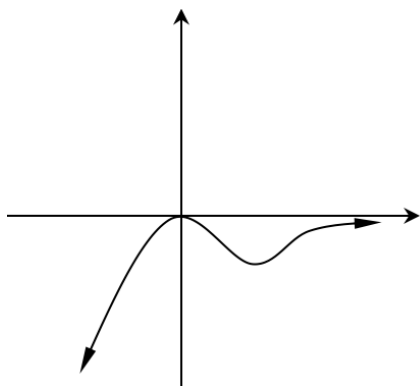
- (A) $V = r\omega$
 (B) $V = mr\omega$
 (C) $a = r\omega^2$
 (D) $a = mr\omega^2$

7. The diagram shows a sketch of the curve $y = xf(x)$.

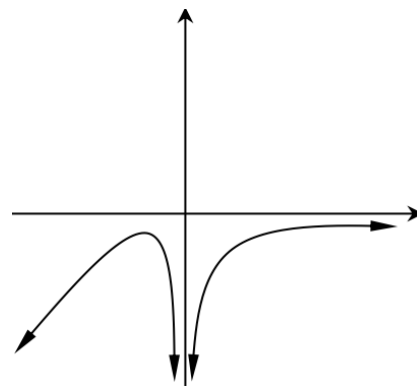


Which of the following best represents $y = f(x)$.

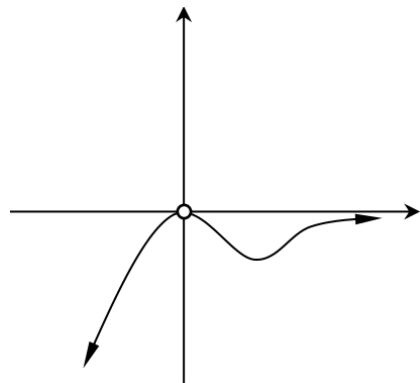
(A)



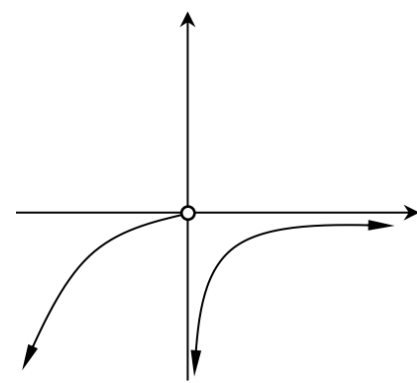
(B)



(C)



(D)



8. Find $\int \cos x \sin^5 x \, dx$

- (A) $\frac{1}{6} \cos^6 x$
- (B) $-\frac{1}{6} \cos x \sin^7 x$
- (C) $6 \sin^6 x$
- (D) $\frac{1}{6} \sin^6 x$

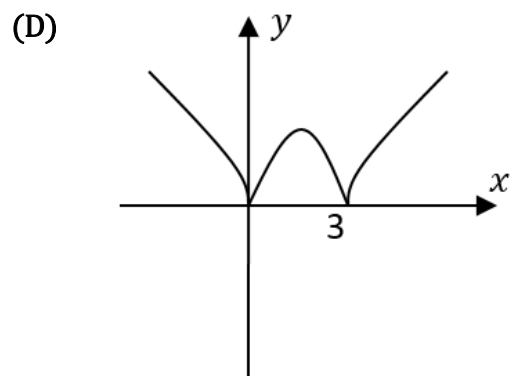
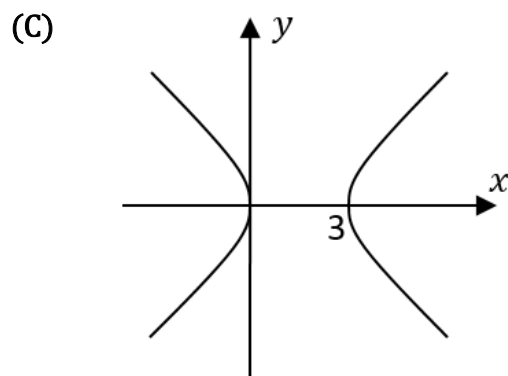
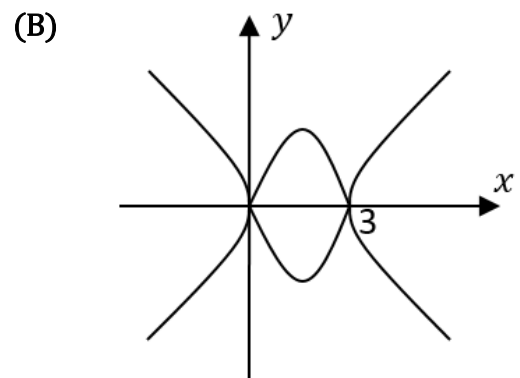
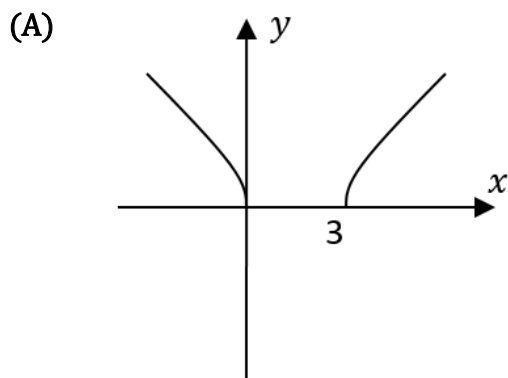
9. A particle of mass m is moving in a straight line under the action of a force

$$F = \frac{m}{x^3}(6 - 10x)$$

Which of the following is an expression for its velocity in any position, if the particle starts from rest at $x = 1$?

- (A) $v = \pm \frac{1}{x} \sqrt{-6 + 5x + x^2}$
 (B) $v = \pm \frac{2}{x} \sqrt{-6 + 5x + x^2}$
 (C) $v = \pm x \sqrt{-6 + 5x + x^2}$
 (D) $v = \pm 2x \sqrt{-6 + 5x + x^2}$

10. If $f(x) = x(x - 3)$, which of the following graphs best represents the curve $y^2 = f(x)$



End of Section I

Section II

Total marks – 90

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Begin each question on a NEW sheet of paper.

Question 11 (15 marks) Use a NEW sheet of paper.

a. Find $\int \frac{dx}{x^2 + 2x + 3}$ **2**

b. Find $\int \cos^5 x \, dx$ **3**

c. Using the substitution $t = \tan \frac{x}{2}$, or otherwise, evaluate **4**

$$\int_0^{\frac{\pi}{4}} \frac{dx}{1 + \cos x}$$

d. Find the equation of the tangent to the curve $2x^2 + 2xy - y^2 + 1 = 0$ at the point $(1, 3)$. **3**

e. Sketch, on separate Argand diagrams, the locus of all points z such that:

i. $\arg z = \frac{\pi}{6}$ **1**

ii. $\arg \bar{z} = \frac{\pi}{6}$ **1**

iii. $\arg(-z) = \frac{\pi}{6}$ **1**

End of Question 11

Question 12 (15 marks) Use a NEW sheet of paper.

a. If $z = 5 - 2i$, find in the form $x + iy$:

i. z^2 **1**

ii. $z + 2\bar{z}$ **1**

iii. $\frac{i}{z}$ **2**

b. The roots of the polynomial equation $2x^3 - 3x^2 + 4x - 5 = 0$ are α, β and γ .
Find the polynomial equation that has roots:

i. $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$ **2**

ii. $2\alpha, 2\beta$ and 2γ **2**

c. Find the exact values of x and y , such that $(x + iy)^2 = 7 - 24i$, where x and y are real. **3**

d.

i. Find real numbers a and b such that **2**

$$\frac{x - 4}{x^2 + 5x + 4} = \frac{a}{x + 4} + \frac{b}{x + 1}$$

ii. Hence evaluate **2**

$$\int_2^4 \frac{x - 4}{x^2 + 5x + 4} dx$$

End of Question 12

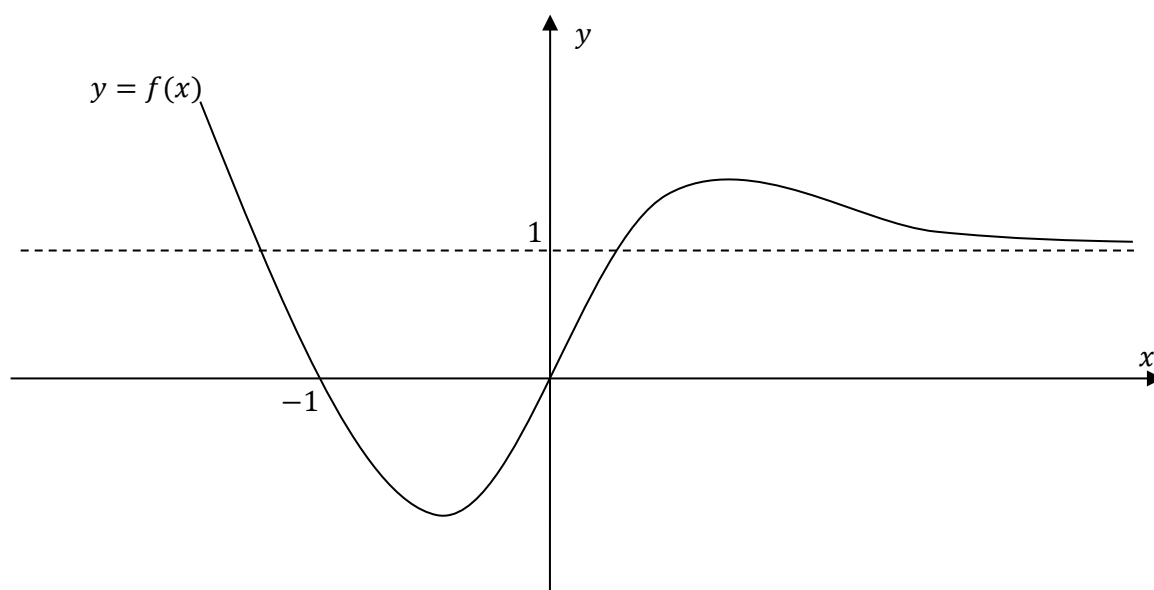
Question 13 (15 marks) Use a NEW sheet of paper.

- a. The base of a solid is the segment of the parabola $x^2 = 4y$ cut off by the line $y = 3$. Cross-sections taken perpendicular to the axis of the parabola are rectangle, whose heights are twice their base.

Find the volume of the solid.

3

- b. The diagram shows the graph of $y = f(x)$.



Draw separate **one-third** page sketches of the graphs of the following:

i. $y = |f(x)|$

1

ii. $y^2 = f(x)$

2

- c. Find the cube roots of $1 - i$.

4

- d. i. $I_n = \int x^n e^{ax} dx$, where a is a constant.

2

Prove that $I_n = \frac{x^n e^{ax}}{a} - \frac{n}{a} I_{n-1}$.

- ii. Hence find the value of $\int_0^1 x^3 e^{2x} dx$.

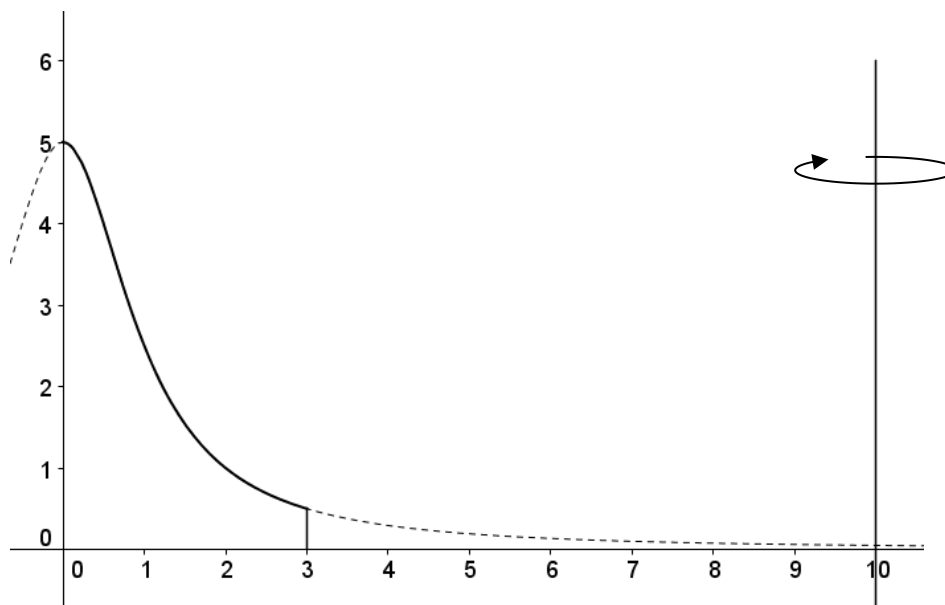
3

End of Question 13

Question 14 (15 marks) Use a NEW sheet of paper.

- a. A circular flange is formed by rotating the region bounded by the curve $y = \frac{5}{x^2 + 1}$, the x -axis and the lines $x = 0$, and $x = 3$, through one complete revolution about the line $x = 10$.

(All measurements are in centimetres)



- i. Use the method of cylindrical shells to show that the volume $V \text{ cm}^3$ of the flange is given by

2

$$V = 10\pi \int_0^3 \frac{10 - x}{x^2 + 1} dx$$

- ii. Find the volume of the flange correct to the nearest cm^3 .

3

- b. $P\left(3p, \frac{3}{p}\right)$ and $Q\left(3q, \frac{3}{q}\right)$ are points on different branches of the hyperbola $xy = 9$.

- i. Find the equation of the tangent at P . 2
- ii. Find the point of intersection, T , of the tangents at P and Q . 2
- iii. If the chord PQ passes through the point $(0, 2)$, find the locus of T . 3
- iv. Find a restriction on the locus of T . 1

- c. On an Argand diagram, sketch the region described by the inequality

2

$$\left|1 + \frac{1}{z}\right| \leq 1.$$

End of Question 14

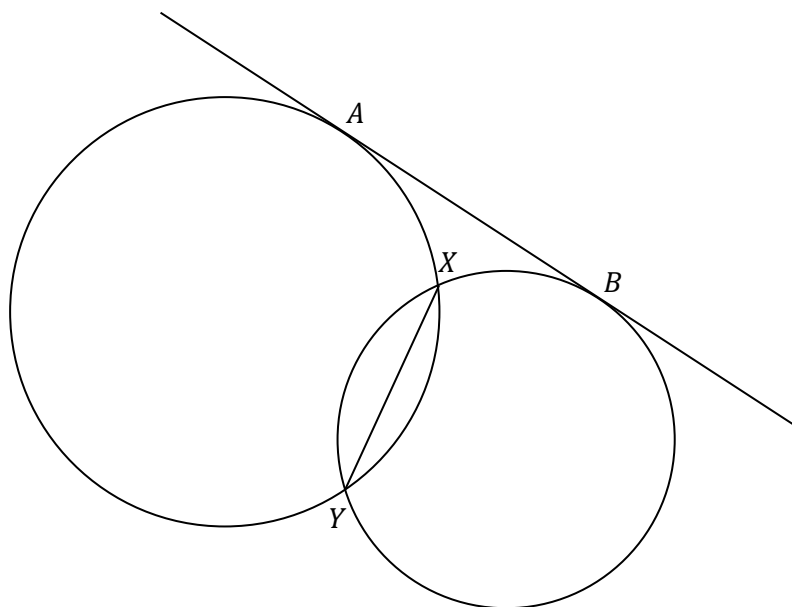
Question 15 (15 marks) Use a NEW sheet of paper.

- a. The polynomial $P(x) = x^5 + 2x^2 + mx + n$ has a double zero at $x = -2$. **3**

Find the product of the other three zeros.

- b. In the diagram below, AB is a common tangent and XY is a common chord to the two circles.

Extend BX to meet AY at Q and extend AX to meet BY at P .



- | | | |
|------|---|----------|
| i. | Copy the diagram onto your answer sheet, showing all the information given. | 1 |
| ii. | Prove that $PXQY$ is a cyclic quadrilateral. | 3 |
| iii. | Prove that AB is parallel to PQ . | 2 |
| iv. | Prove that XY bisects PQ . | 3 |

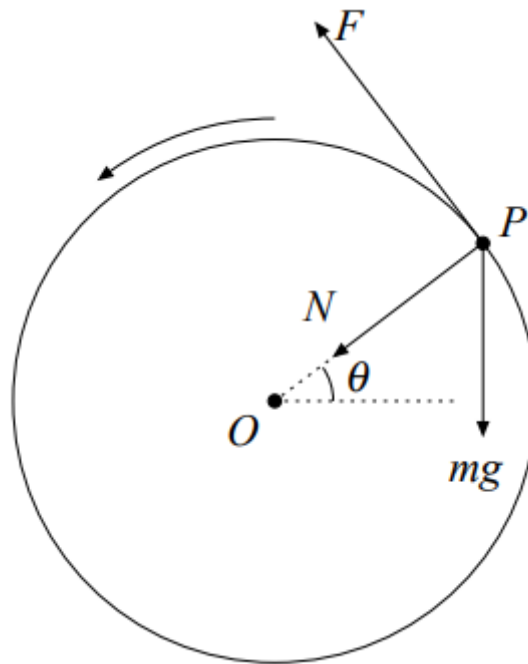
Question 15 is continued on the next page

- c. A circular drum is rotating with uniform angular velocity round a horizontal axis. A particle P is rotating in a vertical circle, without slipping, on the inside of the drum.

The radius of the drum is r metres and its angular velocity is ω radians/second. Acceleration due to gravity is g metres/second², and the mass of P is m kilograms.

The centre of the drum is O , and OP makes an angle θ to the horizontal.

The drum exerts a normal force N on P , as well as a frictional force F , acting tangentially to the drum, as shown in the diagram.



By resolving force perpendicular to, and parallel to, OP , find an expression for $\frac{F}{N}$ in terms of the data.

3

End of Question 15

Question 16 (15 marks) Use a NEW sheet of paper.

- a. A particle is fired vertically upwards with an initial velocity, V metres per second. The particle is subject to air resistance proportional to its speed and a downward gravitational force, g .

The equation of motion of the particle is $\ddot{x} = -g - kv$, where $k > 0$.

Show that the particle reaches a maximum height, H , given by:

5

$$H = \frac{V}{k} - \frac{g}{k^2} \ln \left(1 + \frac{kV}{g} \right)$$

- b. i. Prove that $\frac{1}{2p+1} + \frac{1}{2p+2} > \frac{1}{p+1}$, for all $p > 0$

2

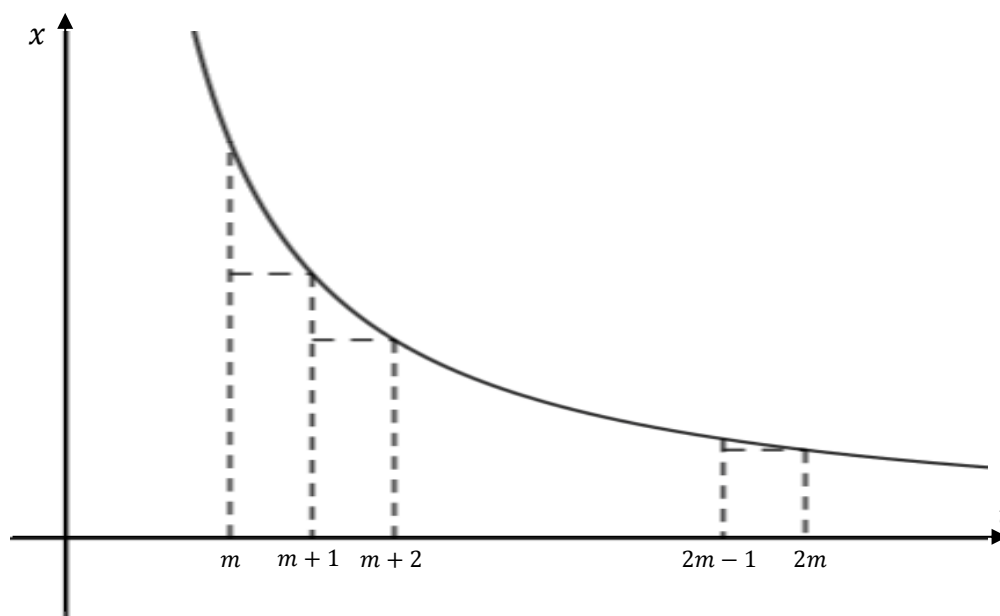
- ii. Consider the statement

$$\psi(m): \frac{1}{(m+1)} + \frac{1}{m+2} + \cdots + \frac{1}{2m} \geq \frac{37}{60}$$

Show, by mathematical induction, that $\psi(m)$ is true for all integers $m \geq 3$.

3

- iii. The diagram below shows the graph of $x = \frac{1}{t}$, for $t > 0$.



By comparing areas, show that $\int_m^{m+1} \frac{1}{t} dt > \frac{1}{m+1}$.

2

- iv. Hence, without using a calculator, show that $\log_e 2 > \frac{37}{60}$.

3

End of paper

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1 C

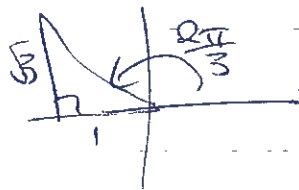
$$w=1, \frac{a}{e}=4.$$

$$a^2=4$$

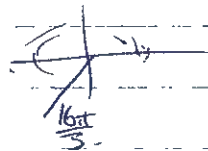
$$w=2 \Rightarrow e=\frac{1}{2} \quad \frac{1}{4}=1-\frac{b^2}{4}$$

$$b^2=3$$

2 C



$$\frac{2\pi}{3} \times 8 = \frac{16\pi}{3}$$



3 C

$$(x-2)^2(x-i)(x+i)$$

4 B

$$\frac{b \sin \theta - b}{a \cos \theta - b} \cdot \frac{b \sin \phi - b}{a \cos \phi - b} = -1$$

$$\tan \theta \cdot \tan \phi = -\frac{a^2}{b^2}$$

5 C/D

$$y = x \ln x$$

$$u = \ln x$$

$$dv = \frac{1}{x} dx$$

$$du = \frac{1}{x} dx$$

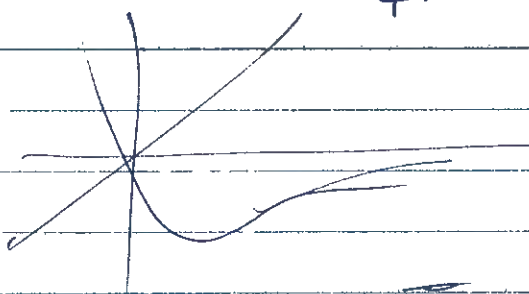
$$v = \frac{x^2}{2}$$

$$I = \frac{x^2}{2} \ln x - \int \frac{x^2}{2x} dx$$

$$= \frac{x^2}{4}$$

6 A

7 C



8

D

$$u = \sin x$$

$$du = \cos x \, dx$$

$$\int u^5 du = \frac{u^6}{6}$$

9.

B.

$$F = \frac{m}{x^3} (6 - 10x)$$

$$u = \frac{6 - 10x}{x^3}$$

$$\int \frac{du}{dx} dx = \int \frac{6 - 10x}{x^3} dx$$

$$\frac{v^2}{2} = \int 6x^{-3} - 10x^{-2} dx$$

$$= \frac{-12}{x^2} + \frac{10}{x} + C$$

$$0 = -12 + 10 + C \Rightarrow C = 2$$

$$\frac{v^2}{2} = \frac{-12}{x^2} + \frac{10}{x} + 2$$

$$v^2 = \frac{2}{x^2} (-12 + 10x + 2x^2)$$

$$= \frac{4}{x^2} (-6 + 5x + x^2)$$

$$v = \pm \frac{2}{x} \sqrt{\dots}$$

10

C

$$11 a \quad I = \int \frac{dx}{x^2 + 2x + 3}$$

$$= \int \frac{dx}{(x+1)^2 + 2}$$

$$= \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x+1}{\sqrt{2}} \right) + C$$

$$b. \quad \int \cos^5 x \, dx = \int \cos^4 x \cos x \, dx$$

$$= \int (1 - \sin^2 x)^2 \cos x \, dx$$

$$u = \sin x$$

$$du = \cos x \, dx$$

$$= \int (1 - u^2)^2 \, du$$

$$= \int (1 - 2u^2 + u^4) \, du$$

$$= u - \frac{2u^3}{3} + \frac{u^5}{5} + C$$

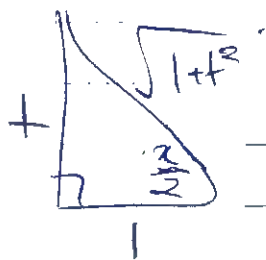
$$= \sin x - \frac{2}{3} \sin^3 x + \frac{1}{5} \sin^5 x + C$$

c.

$$t = \tan \frac{x}{2}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} \, dx$$



$$dx = \frac{2dt \cos^2 \frac{x}{2}}{1+t^2}$$

$$= \frac{2dt}{1+t^2}$$

$$\int_0^{\pi/4} \frac{dx}{1 + \cos x}$$

$$= 2 \int_0^{\tan \pi/8} \frac{1}{1 + \frac{1-t^2}{1+t^2}} \cdot \frac{dt}{1+t^2}$$

$$= 2 \int_0^{\tan \pi/8} \frac{1+t^2}{2} \cdot \frac{dt}{1+t^2}$$

$$= \left[t \right]_0^{\tan \pi/8} = \tan \frac{\pi}{8}$$

$$d. \quad 2x^2 - 2xy - y^2 + 1 = 0$$

$$\frac{d}{dx} (4x - 2y) + 2y - 2y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (2x - 2y) = -(4x + 2y)$$

$$\begin{aligned} \frac{dy}{dx} &= -\frac{(4x + 2y)}{2x - 2y} \\ &= -\frac{(2x + y)}{x - y} \end{aligned}$$

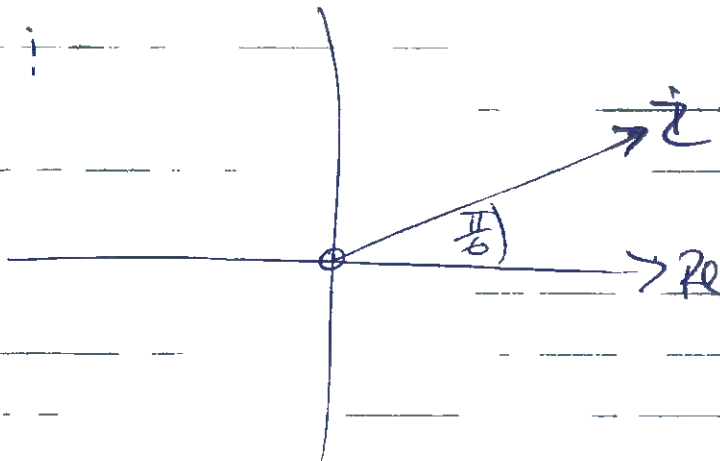
at (1, 3)

$$\begin{aligned} m &= -\frac{(2(1) + 3)}{1 - 3} \\ &= \frac{5}{2} \end{aligned}$$

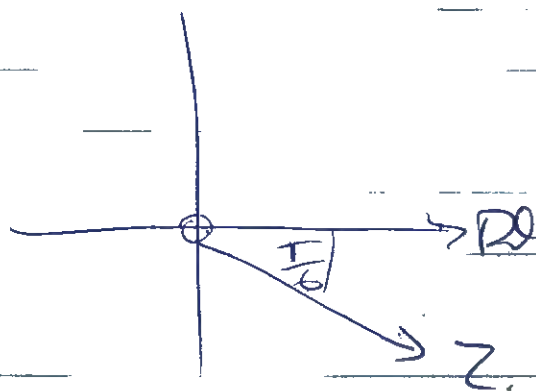
by point-gradient

$$\begin{aligned} y - 3 &= \frac{5}{2}(x - 1) \\ 2y - 6 &= 5x - 5 \\ 5x - 2y + 1 &= 0 \text{ is tangent.} \end{aligned}$$

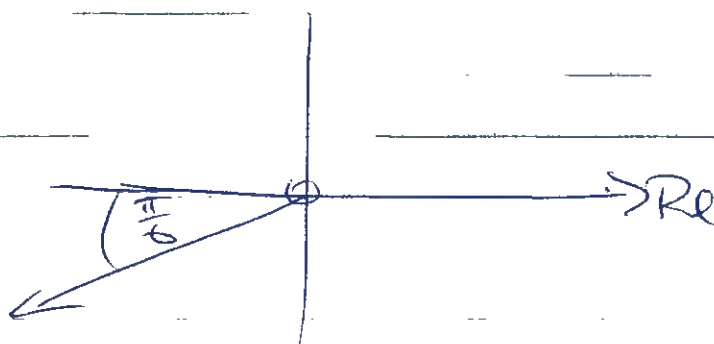
e)



ii



iii



12 a

$$z = 5 - 2i$$

$$\begin{aligned} z^2 &= (5 - 2i)^2 \\ &= 25 - 20i - 4 \\ &= 21 - 20i \end{aligned}$$

$$\begin{aligned} \text{ii. } z + 2\bar{z} &= 5 - 2i + 2(5 + 2i) \\ &= 15 + 2i \end{aligned}$$

$$\begin{aligned} \text{iii. } \frac{1}{z} &= \frac{1}{5 - 2i} \times \frac{5 + 2i}{5 + 2i} \\ &= \frac{-2 + 5i}{25 + 4} \\ &= -\frac{2}{29} + \frac{5i}{29} \end{aligned}$$

12b $2x^3 - 3x^2 + 4x - 5 = 0$

i) $P(\frac{1}{x}): 2(\frac{1}{x})^3 - 3(\frac{1}{x})^2 + 4(\frac{1}{x}) - 5 = 0$

$$\frac{2}{x^3} - \frac{3}{x^2} + \frac{4}{x} - 5 = 0$$

$\times x^3:$ $2 - 3x + 4x^2 - 5x^3 = 0$

$\therefore 5x^3 - 4x^2 + 3x - 2 = 0$ has roots $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$.

ii) $P(\frac{x}{2}): 2(\frac{x}{2})^3 - 3(\frac{x}{2})^2 + 4(\frac{x}{2}) - 5 = 0$

$$\frac{x^3}{4} - \frac{3x^2}{4} + 2x - 5 = 0$$

$\therefore x^3 - 3x^2 + 8x - 20 = 0$ has roots $2x, 2y, 2z$.

c) $(x+iy)^2 = 7-24i$

$$x^2 - y^2 + 2xyi = 7 - 24i$$

equating Re & Im

$$x^2 - y^2 = 7$$

$\times x^2$ $x^4 - x^2y^2 = 7x^2 \quad \dots i$

$$2xy = -24 \quad \dots ii$$

Subst ii into i

$$x^4 - \left(\frac{-24}{2}\right)^2 = 7x^2$$

$$x^4 - 7x^2 - 144 = 0$$

$$(x^2 - 16)(x^2 + 9) = 0$$

$$\therefore x = \pm 4 \quad \text{as } x^2 \neq -9$$

$$y = \frac{-24}{(\pm 4)}$$

$$= \mp 3$$

$\therefore x = 4, y = -3$ and $x = -4, y = 3$ are solutions

d. i. $\frac{x-4}{x^2+5x+4} = \frac{a}{x+4} + \frac{b}{x+1}$

$$\therefore x-4 = a(x+1) + b(x+4)$$

for $x=-1$

$$-5 = a(0) + 3b \Rightarrow b = -\frac{5}{3}$$

for $x=-4$

$$-8 = -3a + b(0) \Rightarrow a = \frac{8}{3}$$

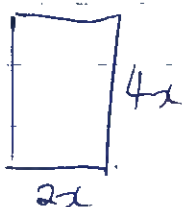
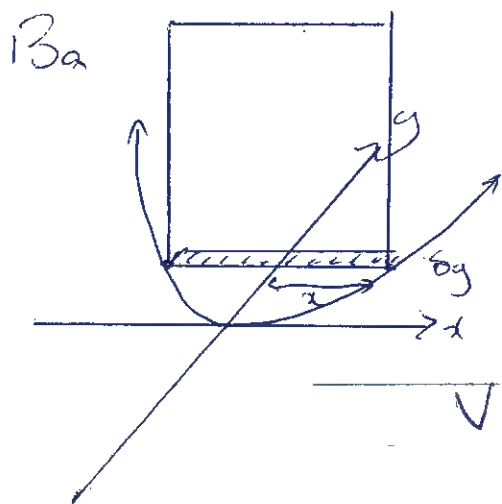
$$\therefore a = \frac{8}{3}, b = -\frac{5}{3}$$

ii. $\int_2^4 \left(\frac{8}{3(x+4)} - \frac{5}{3(x+1)} \right) dx$

$$= \frac{1}{3} \left[8 \ln(x+4) - 5 \ln(x+1) \right]_2^4$$

$$= \frac{1}{3} (8 \ln 8 - 5 \ln 5 - 8 \ln 6 + 5 \ln 3)$$

$$= -0.08422384641$$



$$A = 8x^2$$

$$= 8(4y)$$

$$= 32y$$

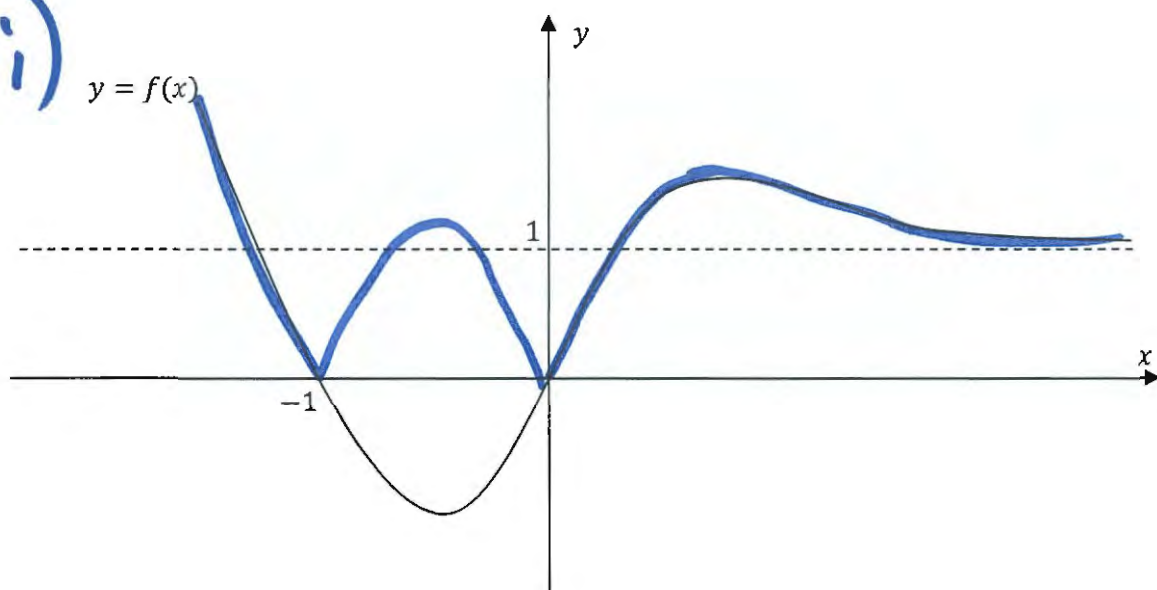
$$\delta V = 32y \cdot \delta y$$

$$V = \lim_{\delta y \rightarrow 0} \sum_{y=0}^3 32y \delta y$$

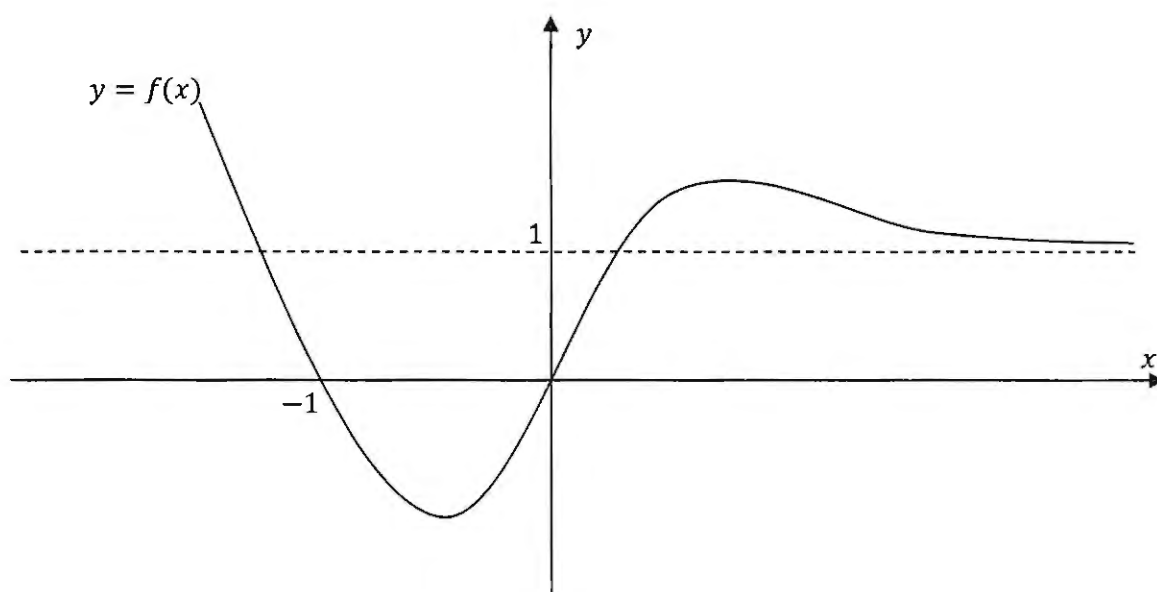
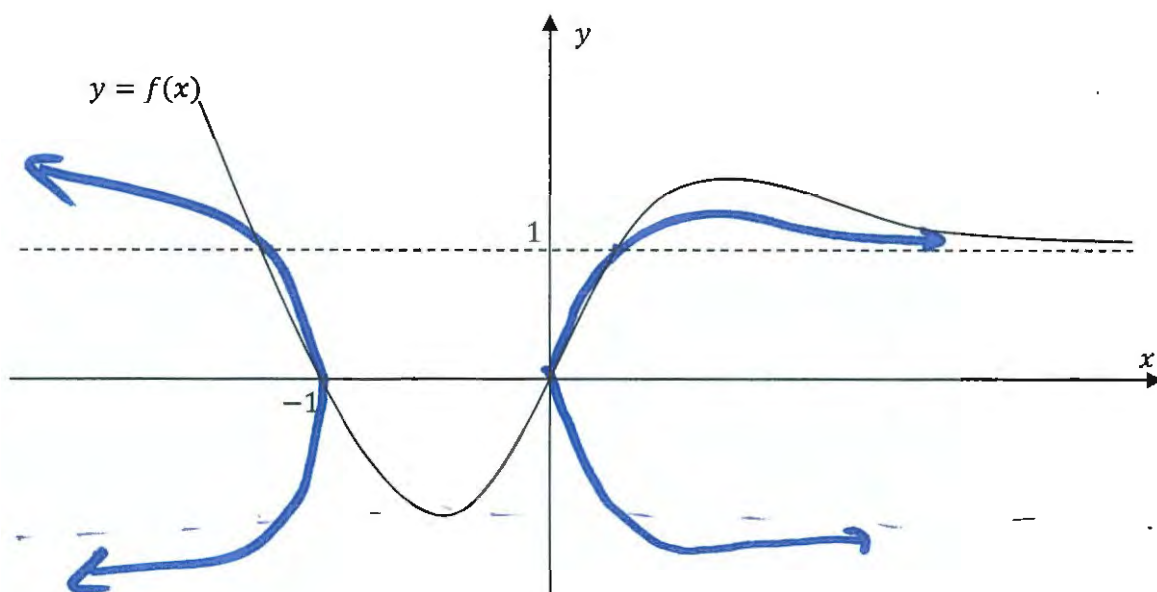
$$= 32 \int_0^3 y dy$$

$$= 32 \left[\frac{y^2}{2} \right]_0^3 = 32 \left(\frac{9}{2} - \frac{0}{2} \right) = 144 \text{ units}^2$$

Bb. i)



ii)



c) $z^3 = 1 - i$
 $\therefore r^3 \operatorname{cis} 3\theta = \sqrt{2} \operatorname{cis}(-\frac{\pi}{4})$

Equating mod + arg

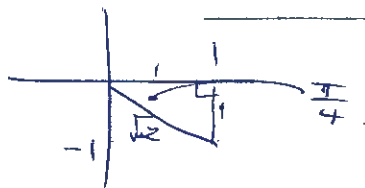
$$r^3 = \sqrt{2}$$

$$r = \sqrt[3]{2}$$

$$\operatorname{cis} 3\theta = \operatorname{cis}(-\frac{\pi}{4})$$

$$3\theta = -\frac{\pi}{4} + 2k\pi$$

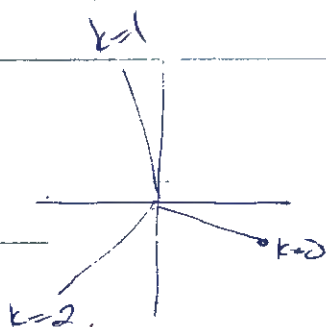
$$\theta = \frac{(8k-1)\pi}{12}$$



for $k=0$, $\theta = -\frac{\pi}{12}$

$k=1$, $\theta = \frac{7\pi}{12}$

$k=2$, $\theta = \frac{15\pi}{12} = -\frac{9\pi}{12}$



$$\therefore z = \sqrt[3]{2} \operatorname{cis}(-\frac{\pi}{12}), \sqrt[3]{2} \operatorname{cis}(\frac{7\pi}{12}), \sqrt[3]{2} \operatorname{cis}(-\frac{9\pi}{12})$$

d) ii

$$I_n = \int x^n e^{ax} dx$$

let $u = x^n$

$$du = nx^{n-1} dx$$

$$dv = e^{ax} dx$$

$$v = \frac{e^{ax}}{a}$$

$$I_n = x^n \frac{e^{ax}}{a} - \int \frac{e^{ax}}{a} \cdot nx^{n-1} dx$$

$$= \frac{x^n e^{ax}}{a} - \frac{n}{a} \int x^{n-1} e^{ax} dx$$

$$= \frac{x^n e^{ax}}{a} - \frac{n}{a} I_{n-1}$$

$$\begin{aligned}
 \text{ii) let } I_3 &= \int_0^1 x^3 e^{2x} dx \\
 &= \left[\frac{x^3 e^{2x}}{2} \right]_0^1 - \frac{3}{2} I_2 \\
 &= \frac{1 \cdot e^{2(1)}}{2} - \frac{0^3 e^{2(0)}}{2} - \frac{3}{2} I_2 \\
 &= \frac{e^2}{2} - \frac{3}{2} I_2
 \end{aligned}$$

$$\begin{aligned}
 I_2 &= \left[\frac{x^2 e^{2x}}{2} \right]_0^1 - \frac{2}{2} I_1 \\
 &= \frac{e^2}{2} - I_1
 \end{aligned}$$

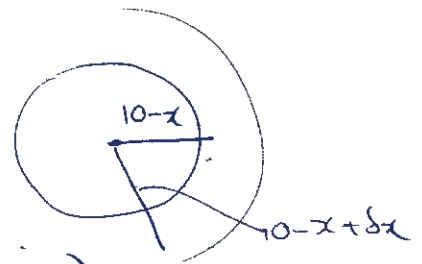
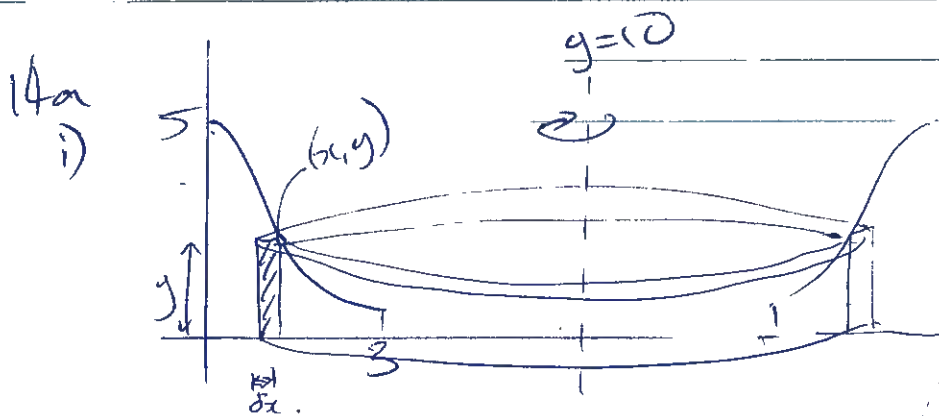
$$\begin{aligned}
 I_1 &= \left[\frac{x e^{2x}}{2} \right]_0^1 - \frac{1}{2} I_0 \\
 &= \frac{e^2}{2} - \frac{1}{2} I_0
 \end{aligned}$$

$$\begin{aligned}
 I_0 &= \int_0^1 e^{2x} dx \\
 &= \left[\frac{e^{2x}}{2} \right]_0^1 \\
 &= \frac{e^2}{2} - \frac{1}{2} \\
 &= \frac{e^2 - 1}{2}
 \end{aligned}$$

$$\begin{aligned}
 \therefore I_1 &= \frac{e^2}{2} - \frac{1}{2} \left(\frac{e^2 - 1}{2} \right) \\
 &= \frac{e^2 + 1}{4}
 \end{aligned}$$

$$\begin{aligned}
 I_2 &= \frac{e^2}{2} - \frac{e^2 + 1}{4} \\
 &= \frac{e^2 - 1}{4}
 \end{aligned}$$

$$\begin{aligned}
 I_3 &= \frac{e^2}{2} - \frac{3}{2} \left(\frac{e^2 - 1}{4} \right) \\
 &= \frac{e^2 + 3}{4}
 \end{aligned}$$



$$\begin{aligned}
 \delta A &= \pi \left((10-x+\delta x)^2 - (10-x)^2 \right) \\
 &= \pi \left((10-x+\delta x - (10-x)) (10-x+\delta x + (10-x)) \right) \\
 &= \pi \delta x (20-x+\delta x) \\
 &= 2\pi (10-x) \delta x \quad \delta x^2 \approx 0
 \end{aligned}$$

$$\begin{aligned}
 \delta V &= 2\pi (10-x) \delta x \cdot y \\
 &= 2\pi (10-x) \delta x \cdot \frac{5}{x^2+1}
 \end{aligned}$$

$$\begin{aligned}
 V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^3 \frac{10\pi (10-x) \delta x}{x^2+1} \\
 &= 10\pi \int_0^3 \frac{10-x}{x^2+1} dx
 \end{aligned}$$

$$\begin{aligned}
 \text{ii)} \quad V &= 100\pi \int_0^3 \frac{dx}{x^2+1} - \frac{10\pi}{2} \int_0^3 \frac{x}{x^2+1} dx \\
 &= 100\pi \left[\tan^{-1} x \right]_0^3 - 5\pi \left[\ln|x^2+1| \right]_0^3 \\
 &= 100\pi \tan^{-1} 3 - 5\pi \ln 10 \\
 &= 356.2303802
 \end{aligned}$$

b) i. $xq = 9$
 $\frac{d}{dx} x \frac{dy}{dx} + y = 0$

$$\frac{dy}{dx} = -\frac{y}{x}$$

at P, $m = -\frac{\frac{3}{p}}{\frac{3p}{p^2}}$
 $= -\frac{1}{p^2}$

by point-gradient

$$y - \frac{3}{p} = -\frac{1}{p^2}(x - 3p)$$

$$p^2 y - 3p = -x + 3p$$

$\therefore x + p^2 y - 6p = 0$ is equation of tangent at P
 — i

ii. Similarly, tangent at Q.

$$x + q^2 y - 6q = 0 \quad \text{— ii}$$

ii-i $(q^2 - p^2)y - (6q - 6p) = 0$
 $y = \frac{6(q-p)}{(q-p)(q+p)}$
 $= \frac{6}{q+p}$

Sub into ii

$$x + q^2 \frac{6}{q+p} - 6q = 0$$

$$x = -\frac{6q^2 + 6q(q+p)}{q+p}$$

$$= -\frac{6pq}{q+p}$$

$\therefore$ Intersection at T is $\left(\frac{6pq}{p+q}, \frac{6}{p+q}\right)$

iii) chord PQ by two-point formula

$$y - \frac{3}{p} = \frac{\frac{3}{q} - \frac{3}{p}}{3q - 3p} (x - 3p)$$
$$= \frac{3(p-q)}{3p(q-p)} (x - 3p)$$

$$y - \frac{3}{p} = -\frac{1}{pq} (x^2 - 3p)$$

$$pqy - 3q = -x^2 + 3p$$

at (0,2)

$$2pq - 3q = -0^2 + 3p$$

$$2pq = 3(p+q)$$

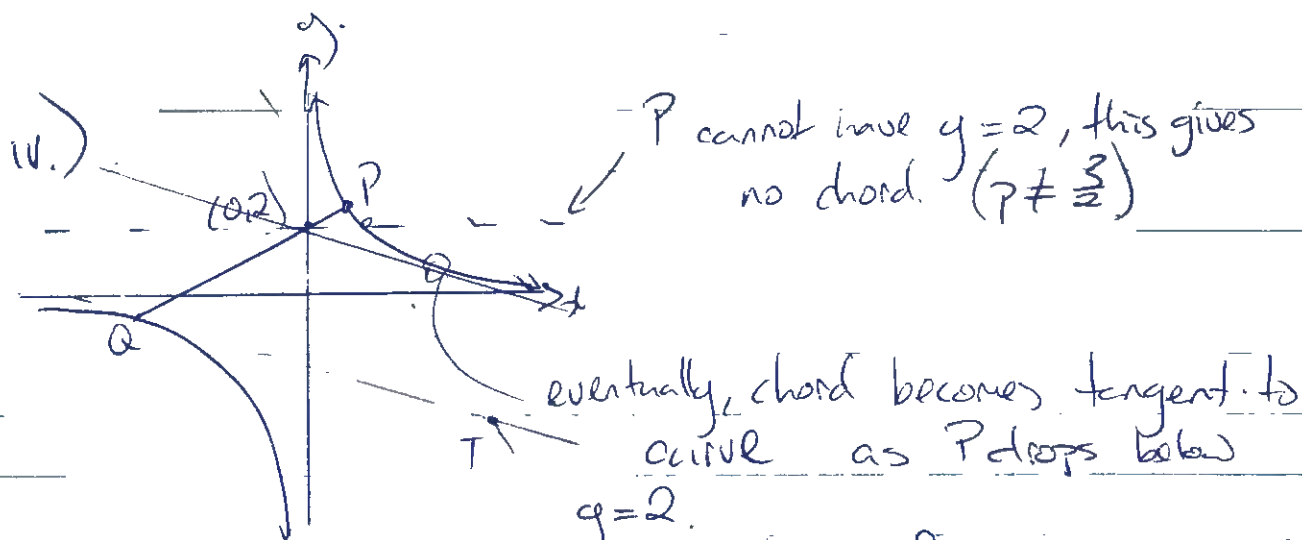
$$\therefore \frac{pq}{p+q} = \frac{3}{2}$$

$\therefore$ Locus of T

$$x = \frac{6pq}{p+q}$$

$$= 6 \left(\frac{3}{2}\right)$$

$$= 9$$



restriction: y -int. of tangent $>$ y -int. of chord.

$$\frac{6p}{p^2} > 2$$

$$\therefore \frac{6}{p} > 2$$

$$\text{hence, } y^P < 3$$

C.

$$\left| 1 + \frac{1}{z} \right| \leq 1$$

$$\left| 1 + \frac{1}{x+iy} \right| \leq 1$$

$$\left| \frac{x+1+iy}{x+iy} \right| \leq 1$$

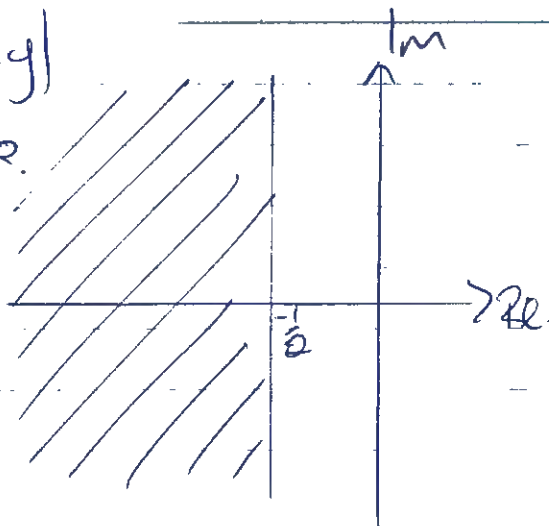
$$|x+1+iy| \leq |x+iy|$$

squaring

$$(x+1)^2 + y^2 \leq x^2 + y^2$$

$$2x+1 \leq 0$$

$$x \leq -\frac{1}{2}$$



Q15

$$P(x) = x^5 + 2x^2 + mx + n$$

$$P(-2) = P'(-2) = 0$$

$$P'(x) = 5x^4 + 4x + m$$

$$P(-2): (-2)^5 + 2(-2)^2 - 2m + n = 0$$

$$\therefore 8 - 2m + n = 0 \quad \text{--- i}$$

$$P'(-2): 5(-2)^4 - 2(4) + m = 0$$

$$\therefore m = -72$$

from i

$$8 - 2(-72) + n = 0$$

$$n = -152$$

$$\text{product of roots: } \alpha\beta\gamma(-2)^2 = \frac{-n}{1}$$

$$4\alpha\beta\gamma = -152$$

$$\therefore \alpha\beta\gamma = -38$$

$\angle ABQ = \angle BQX$ (angle in alternate segment)

Similarly,
 $\angle BAP = \angle APX$

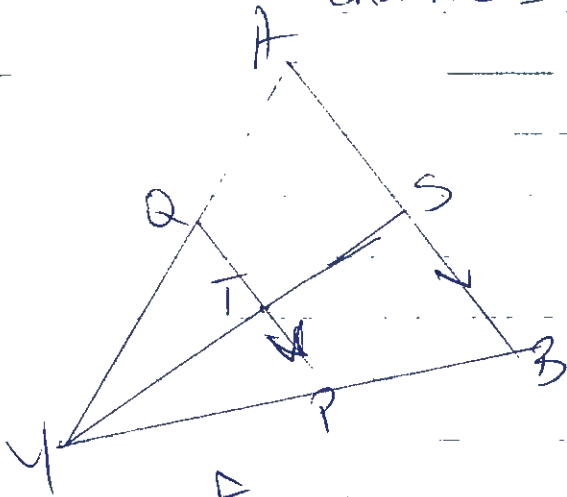
$$\begin{aligned} \angle BAP + \angle ABQ + \angle AYB &= 180^\circ \\ \therefore \angle AYX + \angle BYX + \angle QXP &= 180^\circ \\ \therefore \angle QYP + \angle QXP &= 180^\circ \end{aligned}$$

QXPY is a cyclic quad (opp angles supplementary)

iii) $\angle P, YX = \angle XQP$ (angles in same segment, circle $XQYP$)
 $\therefore \angle ABQ = \angle XQP$
 $PQ \parallel BA$ (alternate angles equal.)

iv) Extend YX to meet AB at S .

$AS^2 = SX \cdot SY$ (square of tangent equals product of secant)
 Similarly
 $BS^2 = SX \cdot SY$
 $\therefore AS^2 = BS^2$
 and $AS = BS$, hence S is the midpoint of AB



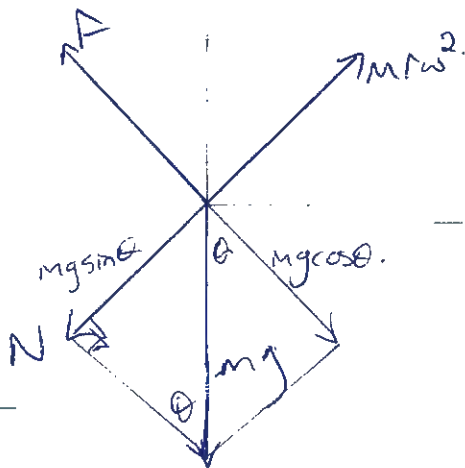
$$\frac{QT}{PT} = \frac{AS}{BS} \quad (\text{ratio of intercepts})$$

$$= 1$$

$$\therefore QT = PT$$

and XY bisects PQ .

c)



perpendicular: $F - mg \cos \theta = 0$

parallel: $N + mg \sin \theta - m r \omega^2 = 0$

$$F = mg \cos \theta$$

$$N = m r \omega^2 - mg \sin \theta$$

$$\frac{F}{N} = \frac{mg \cos \theta}{m(r \omega^2 - g \sin \theta)}$$

$$= \frac{g \cos \theta}{r \omega^2 - g \sin \theta}$$

Q16

a

$$\ddot{x} = -g - kv$$

$$v \frac{dv}{dx} = -g - kv$$

$$\int -\frac{v dv}{g + kv} = \int dx$$

$$x = - \int \frac{v dv}{g + kv}$$

$$= -\frac{1}{k} \int \frac{kv dv}{g + kv}$$

$$= -\frac{1}{k} \int \frac{g + kv - g}{g + kv} dv$$

$$= -\frac{1}{k} \int \left(1 - \frac{g}{g + kv} \right) dv$$

$$= -\frac{1}{k} \left(v - \frac{g}{k} \ln(g + kv) \right) + C$$

$$= -\frac{v}{k} + \frac{g}{k^2} \ln(g + kv) + C$$

for $x=0$, $v=V$

$$C = \frac{V}{k} - \frac{g}{k^2} \ln(g + kV)$$

$$\text{So } x = -\frac{v}{k} + \frac{V}{k} + \frac{g}{k^2} \ln(g + kv) - \frac{g}{k^2} \ln(g + kV)$$

for max height, $v=0$

$$H = \frac{V}{k} + \frac{g}{k^2} \ln(g) - \frac{g}{k^2} \ln(g + kV)$$

$$= \frac{V}{k} - \frac{g}{k^2} \ln \left(\frac{g + kV}{g} \right)$$

$$= \frac{V}{k} - \frac{g}{k^2} \ln \left(1 + \frac{kV}{g} \right)$$

$$b) \quad i) \quad 2^{p+1} < 2^{p+2}$$

$$\frac{1}{2^{p+1}} > \frac{1}{2^{p+2}}$$

$$\frac{1}{2^{p+1}} + \frac{1}{2^{p+2}} > \frac{1}{2^{p+2}} + \frac{1}{2^{p+2}}$$

$$> \frac{2}{2^{p+2}}$$

$$> \frac{1}{2^{p+1}}$$

$$ii) \quad \psi(m): \frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{2m} \geq \frac{37}{60}$$

Prove true for $m=3$

$$LHS = \frac{1}{3+1} + \frac{1}{3+2} + \frac{1}{3+3}$$

$$= \frac{1}{4} + \frac{1}{5} + \frac{1}{6}$$

$$= \frac{37}{60} = RHS \quad \therefore \text{true for } m=3$$

Assume true for $n=k$

$$\therefore \frac{1}{k+1} + \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k} \geq \frac{37}{60}$$

Prove true for $n=k+1$

i.e. - - -

>

$$\text{i.e. } \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k+2} \geq \frac{37}{60}$$

$$\text{LHS} = \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k} + \frac{1}{2k+1} + \frac{1}{2k+2}$$

$$= \frac{1}{k+1} + \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k} + \frac{1}{2k+1} + \frac{1}{2k+2} - \frac{1}{k+1}$$

$$\geq \frac{37}{60} + \frac{1}{2k+1} + \frac{1}{2k+2} - \frac{1}{k+1} \quad \text{by induction hypothesis}$$

$$\geq \frac{37}{60} + \frac{1}{k+1} - \frac{1}{k+1} \quad \text{by part (bi)}$$

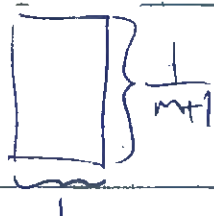
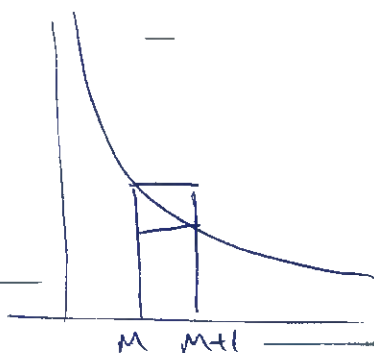
$$\geq \frac{37}{60}$$

$\therefore$ statement is true, by mathematical induction.

iii)

$$\text{low rectangle} < \int_m^{m+1} \frac{1}{t} dt < \text{high rectangle}$$

$$\text{area of low rectangle} = 1 \cdot \frac{1}{m+1}$$



$$\therefore \int_m^{m+1} \frac{1}{t} dt > \frac{1}{m+1}$$

iv)

$$\int_m^{2m} \frac{1}{t} dt = \int_m^{m+1} \frac{1}{t} dt + \int_{m+1}^{m+2} \frac{1}{t} dt + \dots + \int_{2m-1}^{2m} \frac{1}{t} dt$$

$$> \frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{2m} \quad \text{from iii)}$$

$$> \frac{37}{60} \quad \text{from ii)}$$

$$\begin{aligned}\text{Also } \int_m^{2m} \frac{1}{t} dt &= \left[\ln t \right]_m^{2m} \\ &= \ln 2m - \ln m \\ &= \ln \left(\frac{2m}{m} \right) \\ &= \log_e 2\end{aligned}$$

$$\therefore \log_e 2 > \frac{37}{60}$$